28 Here are the first five terms of a Fibonacci sequence.

3 3 6 9 15

(a) Write down the next two terms of the sequence.

..... , ...

(1)

The first three terms of a different Fibonacci sequence are

a a $2a$

(b) Find the 6th term of this sequence.

.....

(2)

(Total for Question 28 is 3 marks)

29 $\mathbf{a} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$ $\mathbf{b} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$

Work out $\mathbf{a} - 2\mathbf{b}$ as a column vector.

$a - 2b = \begin{pmatrix} 4 - 2 \times 3 \\ 5 - 2 \times 2 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$

1 mark
Subtract twice each component of b.

1 mark
Correct column vector.

$\begin{pmatrix} -2 \\ 1 \end{pmatrix}$

(Total for Question 29 is 2 marks)

TOTAL FOR PAPER IS 80 MARKS

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